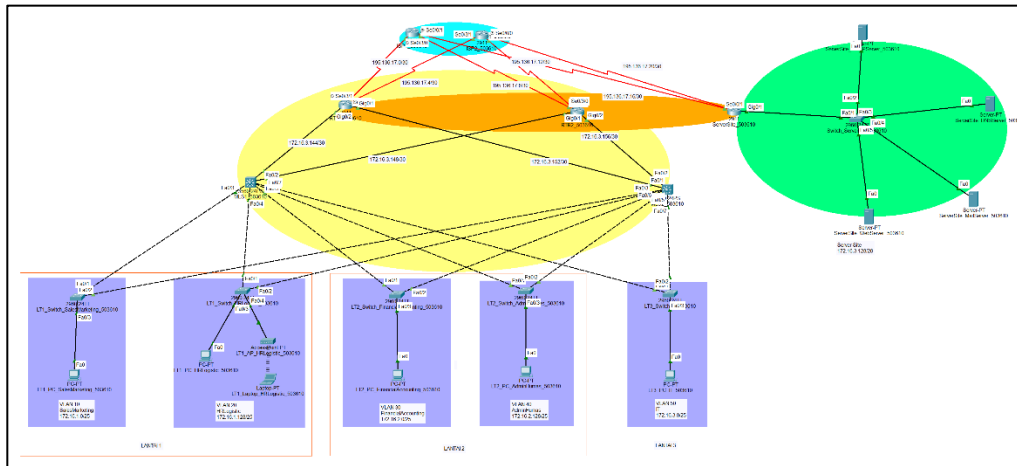


1) Hasil

a) Topologi Sistem

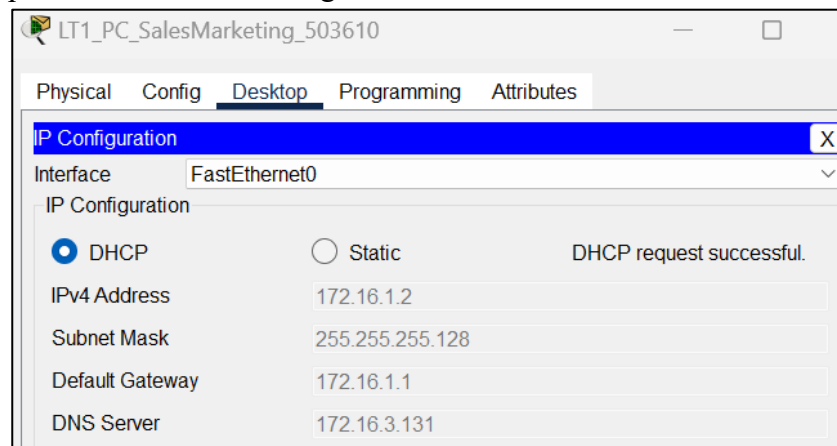


b) Tabel Routing OSPF Server Site: **show ip route ospf**

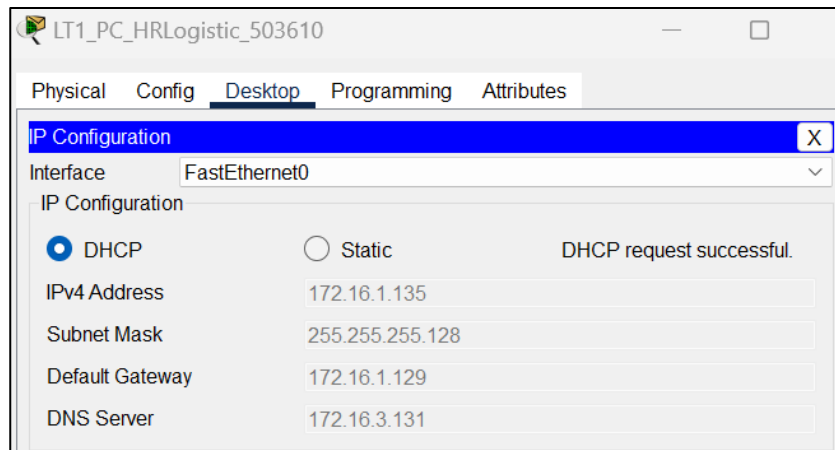
```
ServerSite_503610#sh ip ro ospf
172.16.0.0/16 is variably subnetted, 11 subnets, 4 masks
O 172.16.1.0 [110/130] via 195.136.17.17, 00:02:07, Serial0/0/1
  [110/130] via 195.136.17.21, 00:02:07, Serial0/0/0
O 172.16.1.128 [110/130] via 195.136.17.17, 00:02:07, Serial0/0/1
  [110/130] via 195.136.17.21, 00:02:07, Serial0/0/0
O 172.16.2.0 [110/130] via 195.136.17.17, 00:02:07, Serial0/0/1
  [110/130] via 195.136.17.21, 00:02:07, Serial0/0/0
O 172.16.2.128 [110/130] via 195.136.17.17, 00:02:07, Serial0/0/1
  [110/130] via 195.136.17.21, 00:02:07, Serial0/0/0
O 172.16.3.0 [110/130] via 195.136.17.17, 00:02:07, Serial0/0/1
  [110/130] via 195.136.17.21, 00:02:07, Serial0/0/0
O 172.16.3.144 [110/129] via 195.136.17.17, 00:02:07, Serial0/0/1
  [110/129] via 195.136.17.21, 00:02:07, Serial0/0/0
O 172.16.3.148 [110/129] via 195.136.17.17, 00:02:07, Serial0/0/1
  [110/129] via 195.136.17.21, 00:02:07, Serial0/0/0
O 172.16.3.152 [110/129] via 195.136.17.17, 00:02:07, Serial0/0/1
  [110/129] via 195.136.17.21, 00:02:07, Serial0/0/0
O 172.16.3.156 [110/129] via 195.136.17.17, 00:02:07, Serial0/0/1
  [110/129] via 195.136.17.21, 00:02:07, Serial0/0/0
O 195.136.17.0/24 is variably subnetted, 8 subnets, 2 masks
O 195.136.17.0 [110/128] via 195.136.17.17, 00:02:37, Serial0/0/1
O 195.136.17.4 [110/128] via 195.136.17.21, 00:02:37, Serial0/0/0
O 195.136.17.8 [110/128] via 195.136.17.17, 00:02:37, Serial0/0/1
O 195.136.17.12 [110/128] via 195.136.17.21, 00:02:37, Serial0/0/0
```

c) Hasil DHCP Server pada masing – masing Komputer di Gedung Utama.

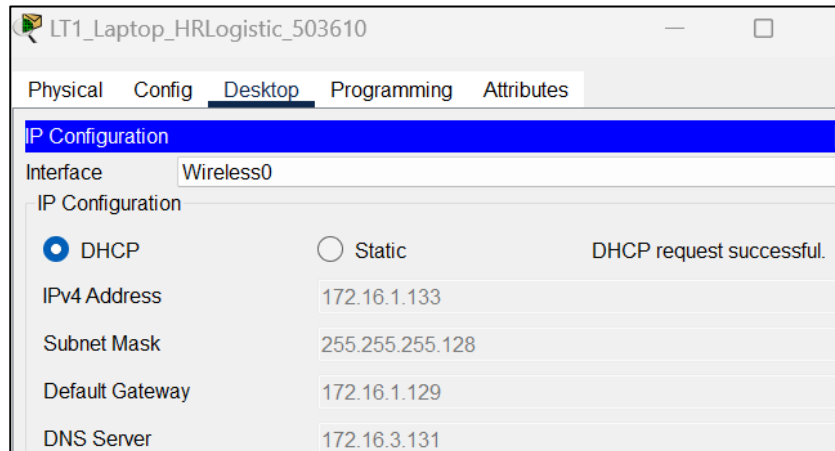
- Komputer Departemen SalesMarketing LT 1



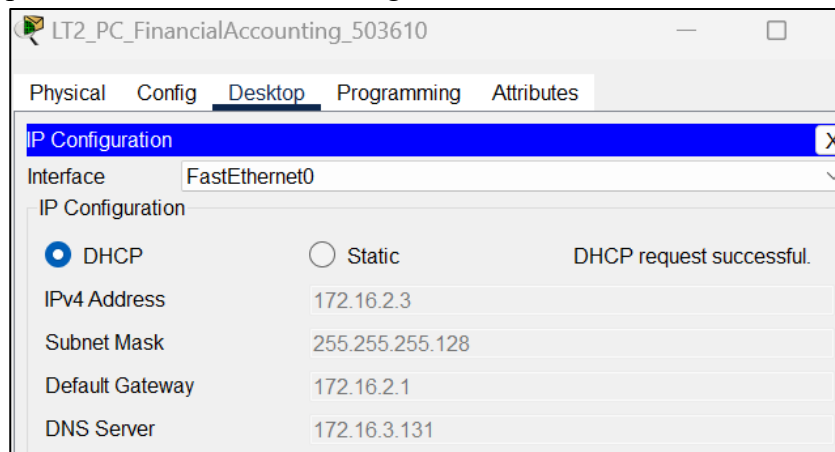
- Komputer Departemen HRLogistic LT 1



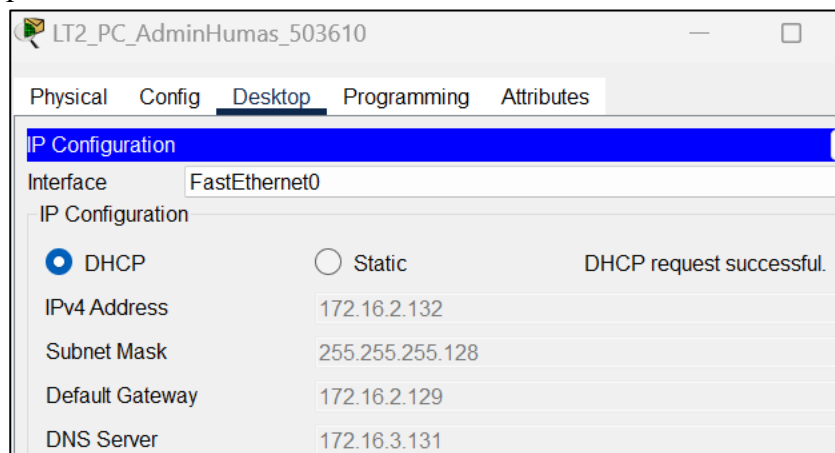
- Laptop Departemen HRLogistic LT 1



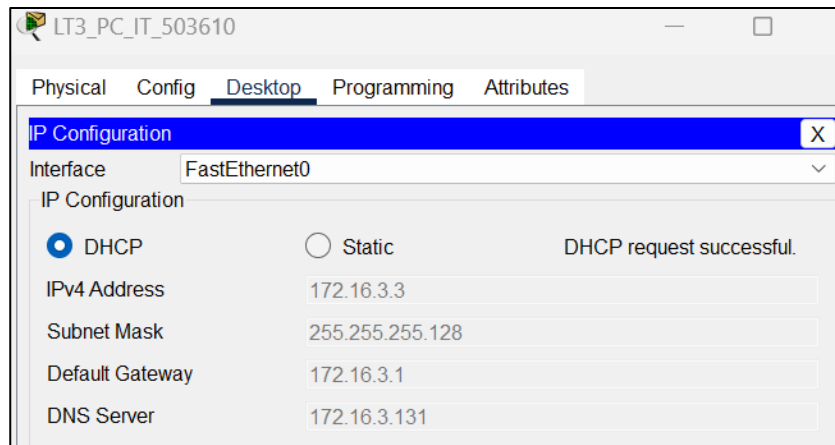
- Komputer Departemen FinancialAccounting LT 2



- Komputer Departemen AdminHumas LT 2

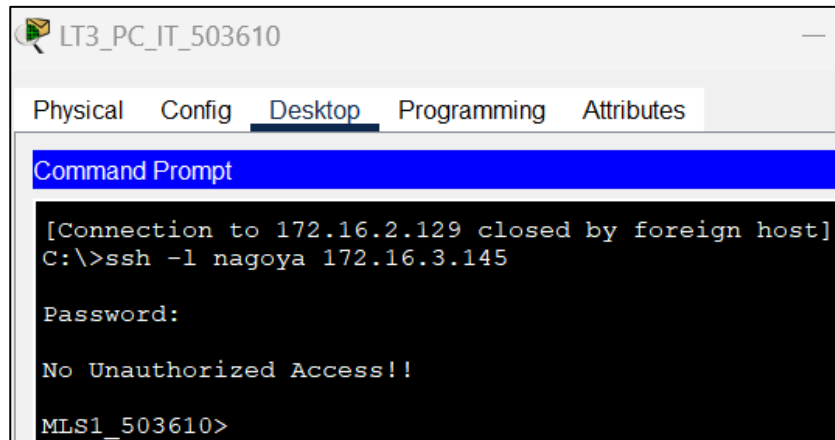


- Komputer Departemen IT LT 3

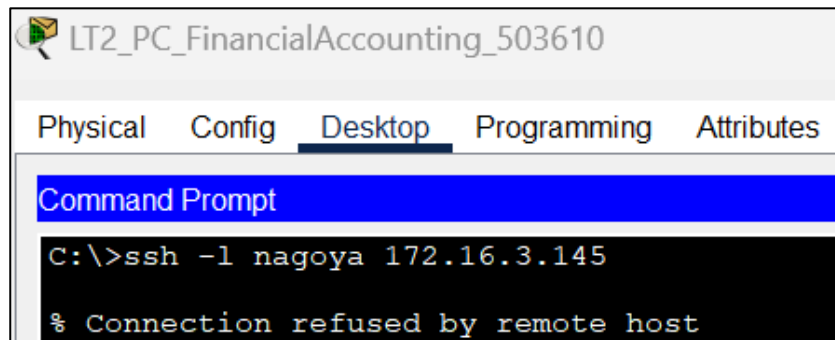


d) Hasil SSH-ACL

- Komputer Departemen IT ke Komputer Departemen AdminHumas

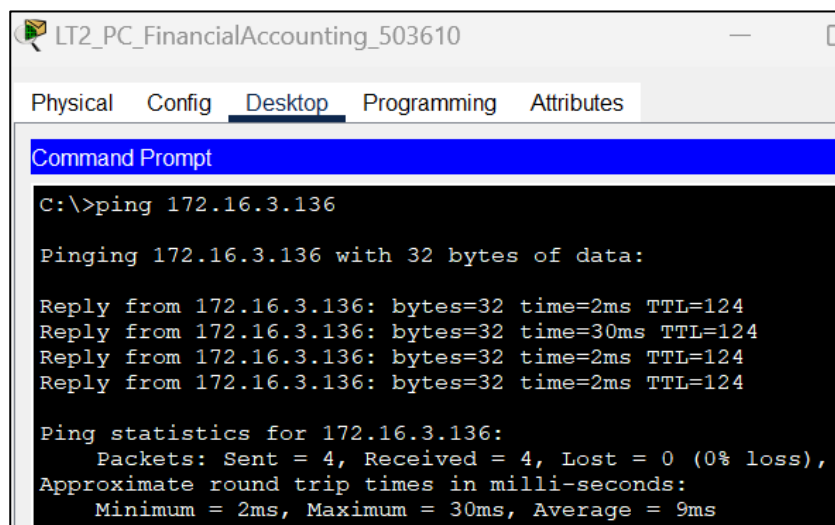


- Komputer Departemen FinancialAccounting ke Komputer Departemen AdminHumas



e) Hasil Uji Coba IPsec-VPN.

- Lakukan PING terlebih dahulu dari salah satu perangkat di Gedung utama ke server di Server Site.



- Hasil dari IPSEC VPN (show crypto ipsec sa dan show crypto isakmp sa)
 - Router 1

```
RTR1_503610
Physical Config CLI Attributes
IOS Command Line Interface
RTR1_503610#sh cry ipsec sa
interface: Serial0/3/0
  Crypto map tag: VPN-MAP, local addr 195.136.17.2

  protected vrf: (none)
  local ident (addr/mask/prot/port):
(172.16.1.0/255.255.255.0/0/0)
  remote ident (addr/mask/prot/port):
(172.16.3.128/255.255.255.240/0/0)
  current_peer 195.136.17.18 port 500
    PERMIT, flags={origin_is_acl,}
    #pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
    #pkts decaps: 13, #pkts decrypt: 13, #pkts verify: 0
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0

  local crypto endpt.: 195.136.17.2, remote crypto endpt.:
195.136.17.18
  path mtu 1500, ip mtu 1500, ip mtu idb Serial0/3/0
  current outbound spi: 0xB92C4367(3106685799)
```

```
RTR1_503610
Physical Config CLI Attributes
IOS Command Line Interface
path mtu 1500, ip mtu 1500, ip mtu idb Serial0/3/0
current outbound spi: 0xB92C4367(3106685799)

inbound esp sas:
  spi: 0x12282727(304621351)
    transform: esp-aes esp-sha-hmac ,
    in use settings = {Tunnel, }
    conn id: 2001, flow_id: FPGA:1, crypto map: VPN-MAP
    sa timing: remaining key lifetime (k/sec):
(4525504/3347)
    IV size: 16 bytes
    replay detection support: N
    Status: ACTIVE

inbound ah sas:

inbound pcsp sas:

outbound esp sas:
  spi: 0xB92C4367(3106685799)
    transform: esp-aes esp-sha-hmac ,
    in use settings = {Tunnel, }
    conn id: 2002, flow_id: FPGA:1, crypto map: VPN-MAP
    sa timing: remaining key lifetime (k/sec):
(4525504/3347)
    IV size: 16 bytes
    replay detection support: N
    Status: ACTIVE

outbound ah sas:

outbound pcsp sas:
```

```
RTR1_503610#sh cry isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id slot
status
195.136.17.18 195.136.17.2  QM_IDLE       1063      0
ACTIVE

IPv6 Crypto ISAKMP SA
```

➤ Server Site

```
ServerSite_503610
Physical Config CLI Attributes
IOS Command Line Interface
ServerSite_503610#sh cry ipsec sa
interface: Serial0/0/1
  Crypto map tag: VPN-MAP, local addr 195.136.17.18

  protected vrf: (none)
  local ident (addr/mask/prot/port):
(172.16.3.128/255.255.255.240/0/0)
  remote ident (addr/mask/prot/port):
(172.16.1.0/255.255.255.0/0/0)
  current_peer 195.136.17.2 port 500
    PERMIT, flags={origin_is_acl,}
    #pkts encaps: 13, #pkts encrypt: 13, #pkts digest: 0
    #pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 1, #recv errors 0

  local crypto endpt.: 195.136.17.18, remote crypto
endpt.:195.136.17.2
  path mtu 1500, ip mtu 1500, ip mtu idb Serial0/0/1
  current outbound spi: 0x12282727(304621351)

inbound esp sas:
  spi: 0xB92C4367(3106685799)
```

```
ServerSite_503610
Physical Config CLI Attributes
IOS Command Line Interface
current outbound spi: 0x12282727(304621351)

inbound esp sas:
  spi: 0xB92C4367(3106685799)
    transform: esp-aes esp-sha-hmac ,
    in use settings ={Tunnel, }
    conn id: 2001, flow_id: FPGA:1, crypto map: VPN-MAP
    sa timing: remaining key lifetime (k/sec):
(4525504/3295)
    IV size: 16 bytes
    replay detection support: N
    Status: ACTIVE

inbound ah sas:

inbound pcsp sas:

outbound esp sas:
  spi: 0x12282727(304621351)
    transform: esp-aes esp-sha-hmac ,
    in use settings ={Tunnel, }
    conn id: 2002, flow_id: FPGA:1, crypto map: VPN-MAP
    sa timing: remaining key lifetime (k/sec):
(4525504/3295)
    IV size: 16 bytes
    replay detection support: N
    Status: ACTIVE

outbound ah sas:

outbound pcsp sas:

ServerSite_503610#sh cry isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id slot
status
195.136.17.2 195.136.17.18 QM_IDLE        1062    0
ACTIVE
```